

ECON 1001 - Midterm 1

2 October 2024

NAME: _____

CANVAS ID: _____

Put your name and your Canvas ID on this cover page and your Scantron sheet.

There are 18 multiple-choice problems and 2 free-response problems. Each problem or subproblem is worth 4 points for a total of 100 points.

For the multiple-choice problems, choose the single best response based on the model we have discussed in class. Fill it in on the Scantron sheet.

For the free-response problems, write your response on the page.

- 1) "People who earn the minimum wage in Washington, DC, do not make enough money to afford the average rent for a 2-bedroom apartment." This statement is
 - a) positive.
 - b) normative.
 - c) transitive.
 - d) [a and b]
- 2) Suppose my willingness to pay for a cookie is higher than your willingness to pay for it. Then we know that
 - a) I get more of a thrill from a cookie than you do.
 - b) I will buy more cookies than you do.
 - c) I have a higher income than you have.
 - d) [We can't be sure that any of the above options are true.]
- 3) Crescent dropped out of medical school before graduating, even though he'd already spent \$200,000 on tuition. He must
 - a) believe he could make more money in some profession that doesn't require a medical degree.
 - b) believe the financial benefit of finishing medical school is smaller than the financial cost.
 - c) think the opportunity cost of staying in medical school is too high.
 - d) be committing the sunk cost fallacy.
- 4) In September, Bobby paid \$200 to reserve a spot for a camping trip in Shenandoah National Park over Thanksgiving weekend (the \$200 is nonrefundable, but there would be no additional financial expenses associated with the trip). However, yesterday an attractive classmate of Bobby's invited Bobby to come to her family home in Pittsburgh for Thanksgiving (there would be no need for Bobby to spend any money if Bobby went to Pittsburgh).
 - a) If Bobby expects to have more than \$200 worth of enjoyment hanging out with this attractive classmate and her family, Bobby will choose to go to Pittsburgh.
 - b) Bobby compares his current WTP for going camping to his WTP for going to Pittsburgh with the classmate, and he chooses the option for which he has the highest WTP.
 - c) It depends on what the price of gas is around Thanksgiving.
 - d) Bobby creates his "true WTP" for camping by adding \$200 to his current WTP for going camping. He compares that to his WTP for going to Pittsburgh, and chooses the best option based on that comparison.
- 5) An individual's current *demand curve* for cookies is probably affected by

- a) the current price of cookies.
 - b) the individual's current income.
 - c) the individual's current level of hunger.
 - d) [b and c]
- 6) During the pandemic, the price of N-95 masks went up, and the quantity of masks purchased increased. That is an example of
- a) an exception to the law of demand.
 - b) positive price elasticity of demand.
 - c) the ceteris paribus assumption not being satisfied.
 - d) [All of the above]
- 7) Demand curves are downward sloping because
- a) as people buy more goods, manufacturers can make the goods for cheaper.
 - b) some stores offer bulk discounts.
 - c) diminishing marginal benefits of consumption.
 - d) [None of the above]
- 8) Suppose tomorrow morning you have two final exams. You have studied a lot already, and now you have just two more hours to study. Each hour of studying for your economics exam will increase your score on that exam by an additional 1 point (compared to if you didn't study any more for it). Each hour of studying for your anthropology exam will increase your score on that exam by an additional 2 points. How many hours of economics should you study?
- a) Zero.
 - b) One.
 - c) Two.
 - d) Can't say without information about how much you value points on the two exams.
- 9) Suppose you are the manager of a shop that sells frozen bananas. Your boss has insisted that you charge \$3 per banana. Assume at that price there is unlimited demand for your bananas and that your marginal cost curve is upward sloping. Because your goal is to maximize the shop's economic surplus (i.e., profits), how many bananas do you sell?
- a) As many as you physically can.
 - b) Keep selling more frozen bananas up to the quantity where total revenue (\$3 times the quantity of frozen bananas you sell) is greater than the total cost.
 - c) You keep selling frozen bananas as long as the cost to you of the final banana you sell is not greater than \$3.
 - d) It depends on the substitutes for frozen bananas that are being sold by other firms in the neighborhood.
- 10) Suppose your friend Tea is at an all-she-can-eat sushi buffet. Tea already paid \$30 for the buffet, which allowed her to eat as much sushi as she wants. How much sushi did Tea eat?
- a) She determined how much sushi she would be willing to pay \$30 for and ate that much.
 - b) She kept eating sushi until her WTP for the last bite of sushi was 0.
 - c) She ate sushi just up to the point where she would be very sick if she ate one more bite.
 - d) She kept eating sushi as long as the MB of an additional piece was greater than 30.

- 11) Coco is the manager of a widget factory. She is considering hiring additional workers (each worker makes additional widgets), but she has noticed a *declining marginal product* of labor from the workers. That means
- a) That each additional worker she hires produces fewer widgets than the previous worker.
 - b) She should hire a new worker only if the wage of the worker is less than that worker's "marginal revenue product" (i.e., the price of a widget times the worker's marginal product).
 - c) Her supply curve is upward sloping (assuming that the marginal costs for other inputs are constant or increasing)
 - d) [All of the above options are true.]

- 12) Suppose the market for widgets in your town is perfectly competitive. Widget makers discover a new way of making widgets that requires fewer inputs. As a result,
- a) the widget makers are able to raise their prices.
 - b) the equilibrium price falls, and more widgets are sold.
 - c) the equilibrium price stays the same, but more widgets are sold.
 - d) demand for widgets increases.

- 13) Rebekah runs a cookie business (this story is familiar from lecture). Each week, at the beginning of the week, before she knows the market price or how many cookies she will sell, she rents a "cookie stand" (fixed cost). In addition to that cost, she also has the variable costs (ingredients, packaging, labor, etc.).

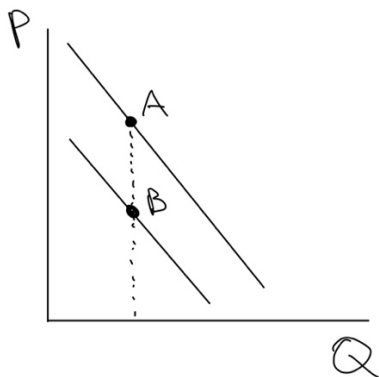
Suppose last week Rebekah rented a cookie stand for \$10 and she sold 15 batches of cookies. But this week the rental cost of the stand fell to \$5. (Suppose the market price for cookies and the prices of other inputs are the same as last week.) Thus, compared to last week, this week

- a) Rebekah will sell the same number of cookies.
 - b) Rebekah will sell more cookies, because now her costs are lower.
 - c) Rebekah will sell fewer cookies, because now her costs are lower.
 - d) Rebekah's total profit will be lower.
- 14) Sparrow is planning a BBQ. She went to the store intending to purchase tofu and some other things. But instead of buying tofu she bought veggie burgers. Assuming that for Sparrow tofu and veggie burgers are substitutes, it could have been the case that
- a) the price of tofu was higher than what Sparrow was expecting.
 - b) the price of a good that was a complement to tofu increased.
 - c) the price of some other good that was a complement to veggie burgers fell.
 - d) [All of the options are potentially valid explanations for Sparrow's behavior.]
- 15) David sells very fine boots from his boot shop. He announced that today at 4pm he will begin selling his newest style at a price of \$500/pair. A line of people started forming at noon and by 5pm there were no new boots remaining. Based on this information, we can be confident that
- a) people who valued the boots the most got them.
 - b) some people who valued the boots at more than \$500 did not get them.
 - c) David has priced the boots on the inelastic portion of the demand curve.
 - d) because of congestion effects, the boots will be worth less than they otherwise might have been.
- 16) The government of Minahstan needs money, so it decided to raise its taxes. But the government knew it would be politically unpopular to announce it was raising taxes on the common people (i.e., the middle and lower class),

so instead it implemented a new tax that the grocery stores had to pay. According to our analysis of taxation from class, as a result of the tax increase:

- a) The common people of Minahstan will be basically unaffected.
 - b) The quantity of goods grocery stores supply will not change.
 - c) Most or all of the economic burden of the tax will be borne by grocery stores.
 - d) The common people of Minahstan will face higher prices.
- 17) Leather goods and beef are complements-in-production. Suppose both markets were in equilibrium and there was an increase in the *demand* for leather goods. Ceteris paribus, that led to
- a) an increase (i.e., outward shift) in the demand for beef.
 - b) there was an increase in the price of beef.
 - c) there was a decrease in the price of beef.
 - d) Cannot say what, if any, effect that had on the price of beef.
- 18) Consider the graph in Figure 1 (next page). Suppose that point B is directly below point A and the two demand curves are parallel. What can you say about the price elasticity of demand at those two points?
- a) Demand is less elastic at point A than at point B.
 - b) Demand is more elastic at point A than at point B.
 - c) Demand is equally elastic at both points.
 - d) Can't say without more information.

Figure 1



NAME: _____

- 19) Consider the demand and supply functions (equations)

$$Q_D = 20 - PQ_S = 2P - 1$$

- a) Draw a supply and demand graph. Label the axes (P and Q), label the intercepts, and label the equilibrium price and quantity.
- b) Suppose the government imposed a price *floor* of 9. How many goods would actually be produced and consumed? [If helpful to you, you may include this on your graph above.]

- c) Instead of a price floor, suppose the government imposed a tax of \$2 per unit *on suppliers*. Who would bear more of the economic burden of the tax, suppliers or demanders (or would the burden be equal)? Explain with one or two sentences. [This problem does not require any additional math.]
- d) The price of inputs changed, and the supply curve shifted to $Q_S = 2P - 4$. Is that an increase or decrease in supply? Explain (one sentence).
- 20) Suppose a widget firm in a perfectly competitive market has the following information

Q	Total Cost
1	2
2	6
3	12
4	20
5	30

- a) If the market price is 9, what would be the firm's economic surplus (i.e., its profit, i.e., what would be the total revenue minus total cost)? Briefly explain. [Assume the firm cannot supply fractional units. E.g, the firm can supply 1 unit or 2 units, but not 1.5 units.]
- b) Complete the table that represents part of the firm's supply curve:

P	Q
2	
4	
6	
8	
10	

- c) Estimate the price elasticity of supply using the points on the supply curve at $P = 2$ and $P = 4$. [You do not need to simplify the arithmetic. If you could not solve the previous part, just use the (incorrect) quantities $Q = 1$ and $Q = 2$.]